UNIT 1

Some Basic Concepts of Chemistry

Chemistry is the science of molecules and their transformations. It is the science not so much of the one hundred elements but of the infinite variety of molecules that may be built from them.

What this unit sets out to do

- appreciate the contribution of India in the development of chemistry, and understand the role of chemistry in different spheres of life
- explain the characteristics of the three states of matter, classify substances into elements, compounds and mixtures, and describe their physical and chemical properties
- use scientific notation, determine significant figures, tell precision from accuracy, define the SI base units and convert quantities from one system of units to another
- explain the laws of chemical combination, appreciate atomic mass, average atomic mass, molecular mass and formula mass, describe the mole and molar mass, calculate mass per cent, determine empirical and molecular formulas, and perform stoichiometric calculations

Where chemistry came from

- science can be viewed as a continuing human effort to systematise knowledge for describing and understanding nature, and the branch of it that studies the preparation, properties, structure and reactions of material substances is called chemistry
- chemistry as we understand it today is not a very old discipline, and it was not studied for its own sake, rather it came up as a result of the search for two interesting things, the philosopher's stone, Paras, which would convert all baser metals such as iron and copper into gold, and the elixir of life, which would grant immortality
- people in ancient India already had the knowledge of many scientific phenomenon much before the advent of modern science and applied it in various walks of life, chemistry developing mainly in the form of alchemy and iatrochemistry during 1300-1600 CE
- modern chemistry took shape in the 18th century Europe, after a few centuries of alchemical traditions which were introduced in Europe by the Arabs

In this unit

1.1 Importance of Chemistry

1.2 Nature of Matter

1.3 Properties and their Measurement

1.4 Uncertainty in Measurement

1.5 Laws of Chemical Combinations

1.6 Dalton's Atomic Theory

1.7 Atomic and Molecular Masses

1.8 Mole Concept and Molar Masses

1.9 Percentage Composition

1.10 Stoichiometry

Development of chemistry, continued

What it was called, and what it covered

- other cultures, especially the Chinese and the Indian, had their own alchemical traditions, and these included much knowledge of chemical processes and techniques
- in ancient India chemistry was called Rasayan Shastra, Rastantra, Ras Kriya or Rasvidya, and it included metallurgy, medicine, and the manufacture of cosmetics, glass and dyes
- systematic excavations at Mohenjodaro in Sindh and Harappa in Punjab prove that the story of the development of chemistry in India is very old, archaeological findings showing that baked bricks were used in construction work, and showing the mass production of pottery, which can be regarded as the earliest chemical process, in which materials were mixed, moulded and subjected to heat by using fire to achieve desirable qualities

Glass, metals and cement

- remains of glazed pottery have been found in Mohenjodaro, and gypsum cement, which contains lime, sand and traces of $CaCO_3$, has been used in the construction work
- Harappans made faience, a sort of glass which was used in ornaments, and they melted and forged a variety of objects from metals such as lead, silver, gold and copper, improving the hardness of copper for making artefacts by using tin and arsenic
- a number of glass objects were found in Maski in South India, 1000-900 BCE, and in Hastinapur and Taxila in North India, 1000-200 BCE, glass and glazes being coloured by the addition of colouring agents like metal oxides
- copper metallurgy in India dates back to the beginning of chalcolithic cultures in the subcontinent, and there are much archeological evidences to support the view that technologies for extraction of copper and iron were developed indigenously

What the texts record

- according to Rigveda, tanning of leather and dying of cotton were practised during 1000-400 BCE, while the golden gloss of the black polished ware of northen India could not be replicated and is still a chemical mystery, these wares indicating the mastery with which kiln temperatures could be controlled, and Kautilya's Arthashastra describing the production of salt from sea
- a vast number of statements and material described in the ancient Vedic literature can be shown to agree with modern scientific findings, and copper utensils, iron, gold, silver ornaments and terracotta discs and painted grey pottery have been found in many archaeological sites in north India
- Sushruta Samhita explains the importance of alkalies, and the Charaka Samhita mentions ancient indians who knew how to prepare sulphuric acid, nitric acid and oxides of copper, tin and zinc, the sulphates of copper, zinc and iron and the carbonates of lead and iron
- Rasopanishada describes the preparation of gunpowder mixture, and Tamil texts also describe the preparation of fireworks using sulphur, charcoal, saltpetre, potassium nitrate, mercury and camphor

Nagarjuna, Chakrapani, and the painted walls

- Nagarjuna was a great Indian scientist, a reputed chemist, an alchemist and a metallurgist, and his work Rasratnakar deals with the formulation of mercury compounds, discussing also methods for the extraction of metals like gold, silver, tin and copper
- a book, Rsarnavam, appeared around 800 CE, and it discusses the uses of various furnaces, ovens and crucibles for different purposes and describes methods by which metals could be identified by flame colour
- Chakrapani discovered mercury sulphide, and the credit for inventing soap also goes to him, he having used mustard oil and some alkalies as ingredients, while Indians began making soaps in the 18th century CE, oil of Eranda and seeds of Mahua plant and calcium carbonate being used for making soap
- the paintings found on the walls of Ajanta and Ellora, which look fresh even after ages, testify to a high level of science achieved in ancient India, and Varähmihir's Brihat Samhita is a sort of encyclopaedia, composed in the sixth century CE, which informs about the preparation of glutinous material to be applied on walls and roofs of houses and temples, prepared entirely from extracts of various plants, fruits, seeds and barks, concentrated by boiling and then treated with various resins

Development of chemistry, continued

Colour, scent, ink and drink

- a number of classical texts, like Atharvaveda, 1000 BCE, mention some dye stuff, the material used being turmeric, madder, sunflower, orpiment, cochineal and lac, while some other substances having tinting property were kamplcica, pattanga and jatuka
- Varähmihir's Brihat Samhita gives references to perfumes and cosmetics, recipes for hair dying being made from plants like indigo and minerals like iron power, black iron or steel and acidic extracts of sour rice gruel, while Gandhayukli describes recipes for making scents, mouth perfumes, bath powders, incense and talcum power
- paper was known to India in the 17th century as the account of the Chinese traveller I-tsing describes, and excavations at Taxila indicate that ink was used in India from the fourth century, colours of ink being made from chalk, red lead and minimum
- it seems that the process of fermentation was well-known to Indians, Vedas and Kautilya's Arthashastra mentioning about many types of liquors, and Charaka Samhita also mentioning ingredients such as barks of plants, stem, flowers, leaves, woods, cereals, fruits and sugarcane

Acharya Kanda and the Paramãnu

- the concept that matter is ultimately made of indivisible building blocks appeared in India a few centuries BCE as a part of philosophical speculations
- Acharya Kanda, born in 600 BCE and originally known by the name Kashyap, was the first proponent of the atomic theory, formulated the theory of very small indivisible particles which he named Paramãnu, comparable to atoms, and authored the text Vaiseshika Sutras
- according to him all substances are aggregated form of smaller units called atoms, which are eternal, indestructible, spherical, suprasensible and in motion in the original state, and he explained that this individual entity cannot be sensed through any human organ
- Kanda added that there are varieties of atoms that are as different as the different classes of substances, said these could form pairs or triplets among other combinations and that unseen forces cause interaction between them, conceptualising this theory around 2500 years before John Dalton, 1766-1844

Charaka Samhita and the bhasma

- Charaka Samhita is the oldest Ayurvedic epic of India and it describes the treatment of diseases
- the concept of reduction of particle size of metals is clearly discussed in it, and extreme reduction of particle size is termed as nanotechnology
- it describes the use of bhasma of metals in the treatment of ailments, and now-a-days it has been proved that bhasmas have nanoparticles of metals

The decline, and what chemistry is

- after the decline of alchemy, iatrochemistry reached a steady state, but it too declined due to the introduction and practise of western medicinal system in the 20th century, and during this period of stagnation the pharmaceutical industry based on Ayurveda continued to exist but it too declined gradually
- it took about 100-150 years for Indians to learn and adopt new techniques, during which time foreign products poured in and indigenous traditional techniques gradually declined, modern science appearing in the Indian scene in the later part of the nineteenth century, when European scientists started coming to India and modern chemistry started growing
- chemistry deals with the composition, structure, properties and interection of matter and is of much use to human beings in daily life, and these aspects can be best described and understood in terms of the basic constituents of matter that are atoms and molecules, which is why chemistry is also called the science of atoms and molecules
- can we see, weigh and perceive these entities, is it possible to count the number of atoms and molecules in a given mass of matter and have a quantitative relationship between the mass and the number of these particles, and how can physical properties be described using numerical values with suitable units, are the questions this unit answers

1.1 Importance of chemistry

Where its principles apply

- chemistry plays a central role in science and is often intertwined with other branches of science, and its principles are applicable in diverse areas such as weather patterns, the functioning of brain and the operation of a computer, production in chemical industries, and the manufacture of fertilisers, alkalis, acids, salts, dyes, polymers, drugs, soaps, detergents, metals and alloys, including new material
- chemistry contributes in a big way to the national economy, and it also plays an important role in meeting human needs for food, healthcare products and other material aimed at improving the quality of life, exemplified by the large-scale production of a variety of fertilisers and improved variety of pesticides and insecticides
- it provides methods for the isolation of lifesaving drugs from natural sources and makes possible synthesis of such drugs, some of these being cisplatin and taxol, which are effective in cancer therapy, while the drug AZT, azidothymidine, is used for helping AIDS patients

New materials, and the environment

- with a better understanding of chemical principles it has now become possible to design and synthesise new material having specific magnetic, electric and optical properties, which has lead to the production of superconducting ceramics, conducting polymers and optical fibres, while chemistry has helped in establishing industries which manufacture utility goods like acids, alkalies, dyes, polymesr and metals, contributing in a big way to the economy of a nation and generating employment
- in recent years chemistry has helped in dealing with some of the pressing aspects of environmental degradation with a fair degree of success, safer alternatives to environmentally hazardous refrigerants like the CFCs, chlorofluorocarbons, responsible for ozone depletion in the stratosphere, having been successfully synthesised
- many big environmental problems continue to be matters of grave concern to the chemists, one such problem being the management of the Green House gases like methane and carbon dioxide, while understanding of biochemical processes, use of enzymes for large-scale production of chemicals and synthesis of new exotic material are some of the intellectual challenges for the future generation of chemists, for which a developing country like India needs talented and creative chemists

1.2 Nature of matter, and 1.2.1 states of matter

Matter, and its three states

- matter is anything which has mass and occupies space, and everything around us, book, pen, pencil, water, air, all living beings, is composed of matter
- matter can exist in three physical states viz. solid, liquid and gas, and the constituent particles of matter in these three states can be represented as shown in the figure at the top of the next page
- particles are held very close to each other in solids in an orderly fashion and there is not much freedom of movement, in liquids the particles are close to each other but they can move around, and in gases the particles are far apart as compared to those present in solid or liquid states and their movement is easy and fast
- because of such arrangement of particles the different states of matter exhibit these characteristics, solids have definite volume and definite shape, while liquids have definite volume but do not have definite shape and take the shape of the container in which they are placed

Solid, particles packed close in an orderly way with little freedom of movement

Liquid, particles still close together but able to move around one another

Gas, particles far apart, movement easy and fast

Figure 1.A Arrangement of particles in solid, liquid and gaseous state

Matter, divided first into mixtures and pure substances

Mixtures, either homogeneous with a uniform composition or heterogeneous without one

Pure substances, either elements of one kind of atom or compounds of two or more

Figure 1.B Classification of matter

Gases, and interconversion

- gases have neither definite volume nor definite shape, and they completely occupy the space in the container in which they are placed
- these three states of matter, solid, liquid and gas, are interconvertible by changing the conditions of temperature and pressure
- on heating a solid usually changes to a liquid, and the liquid on further heating changes to gas or vapour, while in the reverse process a gas on cooling liquifies to the liquid and the liquid on further cooling freezes to the solid

1.2.2 Classification of matter

Mixtures

- at the macroscopic or bulk level matter can be classified as mixture or pure substance, and when all constituent particles of a substance are same in chemical nature it is said to be a pure substance, while a mixture contains many types of particles
- a mixture contains particles of two or more pure substances which may be present in it in any ratio, hence their composition is variable, and the pure substances forming a mixture are called its components, many of the substances present around us, such as sugar solution in water, air and tea, being mixtures
- in a homogeneous mixture the components completely mix with each other, so particles of components are uniformly distributed throughout the bulk and its composition is uniform throughout, sugar solution and air being the examples, whereas in a heterogeneous mixture the composition is not uniform and sometimes different components are visible, mixtures of salt and sugar, and grains and pulses along with some dirt, being heterogeneous mixtures
- the components of a mixture can be separated by using physical methods such as simple hand-picking, filtration, crystallisation and distillation

Pure substances, and their two kinds

- constituent particles of pure substances have fixed composition, copper, silver, gold, water and glucose being some examples, and glucose contains carbon, hydrogen and oxygen in a fixed ratio with its particles of same composition, so like all other pure substances it has a fixed composition, and its constituents cannot be separated by simple physical methods
- pure substances can further be classified into elements and compounds, particles of an element consisting of only one type of atoms, which may exist as atoms or molecules

Atoms of different elements, each drawn as its own kind of sphere

Two hydrogen atoms joining to give one molecule of hydrogen

Two oxygen atoms joining to give one molecule of oxygen

Figure 1.C A representation of atoms and molecules

Water, one oxygen atom holding two hydrogen atoms at an angle

Carbon dioxide, one carbon atom with an oxygen atom on either side in a line

Figure 1.D A depiction of molecules of water and carbon dioxide

Atoms and molecules of elements

- sodium, copper, silver, hydrogen and oxygen are some examples of elements, all their atoms being of one type, whereas the atoms of different elements are different in nature
- some elements such as sodium or copper contain atoms as their constituent particles, whereas in some others the constituent particles are molecules which are formed by two or more atoms, so hydrogen, nitrogen and oxygen gases consist of molecules in which two atoms combine to give their respective molecules

Compounds

- when two or more atoms of different elements combine together in a definite ratio the molecule of a compound is obtained, and the constituents of a compound cannot be separated into simpler substances by physical methods, they can be separated by chemical methods, examples of some compounds being water, ammonia, carbon dioxide and sugar
- a water molecule comprises two hydrogen atoms and one oxygen atom, and a molecule of carbon dioxide contains two oxygen atoms combined with one carbon atom, so the atoms of different elements are present in a compound in a fixed and definite ratio which is characteristic of a particular compound
- the properties of a compound are different from those of its constituent elements, so hydrogen and oxygen are gases whereas the compound formed by their combination, water, is a liquid, and it is interesting to note that hydrogen burns with a pop sound and oxygen is a supporter of combustion, but water is used as a fire extinguisher

1.3 Properties of matter, and 1.3.1 physical and chemical properties

The two categories of property

- every substance has unique or characteristic properties, and these can be classified into two categories, physical properties such as colour, odour, melting point, boiling point and density, and chemical properties like composition, combustibility and ractivity with acids and bases
- physical properties can be measured or observed without changing the identity or the composition of the substance, whereas the measurement or observation of chemical properties requires a chemical change to occur, and measurement of physical properties does not require occurance of a chemical change
- the examples of chemical properties are characteristic reactions of different substances, and these include acidity or basicity and combustibility, while chemists describe, interpret and predict the behaviour of substances on the basis of knowledge of their physical and chemical properties, which are determined by careful measurement and experimentation

1.3.2 Measurement of physical properties

A number, and a unit

- quantitative measurement of properties is reaquired for scientific investigation, and many properties of matter such as length, area and volume are quantitative in nature
- any quantitative observation or measurement is represented by a number followed by units in which it is measured, so the length of a room can be represented as 6 m, where 6 is the number and m denotes metre, the unit in which the length is measured
- earlier two different systems of measurement, the English System and the Metric System, were being used in different parts of the world, the metric system, which originated in France in late eighteenth century, being more convenient as it was based on the decimal system, and later the need of a common standard system was felt by the scientific community, such a system being established in 1960

Maintaining the national standards

- the system of units, including unit definitions, keeps on changing with time, and whenever the accuracy of measurement of a particular unit was enhanced substantially by adopting new principles, member nations of the metre treaty, signed in 1875, agreed to change the formal definition of that unit
- each modern industrialised country, including India, has a National Metrology Institute, NMI, which maintains standards of measurements, and this responsibility has been given to the National Physical Laboratory, NPL, New Delhi
- that laboratory establishes experiments to realise the base units and derived units of measurement and maintains National Standards of Measurement, and these standards are periodically inter-compared with standards maintained at other National Metrology Institutes in the world, as well as those established at the International Bureau of Standards in Paris

1.3.3 The International System of Units, SI

Where the SI came from

- the International System of Units, in French Le Systeme International d'Unités and abbreviated as SI, was established by the 11th General Conference on Weights and Measures, CGPM from Conference Generale des Poids et Measures
- the CGPM is an inter-governmental treaty organisation created by a diplomatic treaty known as Metre Convention, which was signed in Paris in 1875
- the SI system has seven base units, which pertain to the seven fundamental scientific quantities, and the other physical quantities such as speed, volume and density can be derived from these quantities

Table 1.A Base Physical Quantities and their Units

Base Physical Quantity	Symbol for Quantity	Name of SI Unit	Symbol for SI Unit
Length	l	metre	m
Mass	m	kilogram	kg
Time	t	second	s
Electric current	I	ampere	A
Thermodynamic temperature	T	kelvin	K
Amount of substance	n	mole	mol
Luminous intensity	Iv	candela	cd

1.3.3 What fixes each base unit

How the definitions work now

- each base unit is now pinned to a constant of nature whose numerical value is fixed by agreement, so the unit follows from the constant rather than from any object
- the definitions of the SI base units are given in the table below, and the SI system also allows the use of prefixes to indicate the multiples or submultiples of a unit, which are listed on the next page

Table 1.8 Definitions of SI Base Units

Unit of	Name	The constant whose fixed value defines it
Length	metre	the speed of light in vacuum, c, fixed at 299792458 in the unit ms–1, with the second taken from the caesium frequency
Mass	kilogram	the Planck constant h, fixed at 6.62607015×10–34 in the unit Js, that is kgm2s–1, with the metre and second taken from c and the caesium frequency
Time	second	the caesium frequency, the unperturbed ground-state hyperfine transition frequency of the caesium-133 atom, fixed at 9192631770 in the unit Hz, that is s–1
Electric current	ampere	the elementary charge e, fixed at 1.602176634×10–19 in the unit C, that is As, with the second taken from the caesium frequency
Thermodynamic temperature	kelvin	the Boltzmann constant k, fixed at 1.380649×10–23 in the unit JK–1, that is kgm2s–2k–1, with the kilogram, metre and second taken from h, c and the caesium frequency
Amount of substance	mole	a fixed count, one mole containing exactly 6.02214076×1023 elementary entities, which is the Avogadro number when expressed in the unit mol–1
Luminous intensity	candela	the luminous efficacy of monochromatic radiation of frequency 540×1012 Hz, fixed at 683 in the unit lm·W–1, that is cd·sr·W–1

An enclosed pan balance, the instrument that fixes a mass accurately in the laboratory

Read in grams, the fraction of the kilogram that laboratory quantities call for

Figure 1.E Analytical balance

1 L is 1000 mL

1000 cm3 is 1 dm3

so a litre and a cubic decimetre are one and the same volume

Figure 1.F Different units used to express volume

1.3.4 Mass and weight, and 1.3.5 volume

Mass is not weight

- mass of a substance is the amount of matter present in it, while weight is the force exerted by gravity on an object, so the mass of a substance is constant whereas its weight may vary from one place to another due to change in gravity, and you should be careful in using these terms
- the mass of a substance can be determined accurately in the laboratory by using an analytical balance, and although the SI unit of mass is kilogram, its fraction named as gram, 1 kg = 1000 g, is used in laboratories due to the smaller amounts of chemicals used in chemical reactions

Volume, and the litre

- volume is the amont of space occupied by a substance and it has the units of length cubed, so in the SI system volume has units of m3, but again in chemistry laboratories smaller volumes are used, hence volume is often denoted in cm3 or dm3 units
- a common unit, litre, L, which is not an SI unit, is used for measurement of volume of liquids, and 1 L = 1000 mL, 1000 cm3 = 1 dm3

Table 1.C Prefixes used in the SI System

Multiple	Prefix	Symbol	Multiple	Prefix	Symbol
10–24	yocto	y	10	deca	da
10–21	zepto	z	102	hecto	h
10–18	atto	a	103	kilo	k
10–15	femto	f	106	mega	M
10–12	pico	p	109	giga	G
10–9	nano	n	1012	tera	T
10–6	micro	μ	1015	peta	P
10–3	milli	m	1018	exa	E
10–2	centi	c	1021	zeta	Z
10–1	deci	d	1024	yotta	Y

Graduated cylinder, burette and pipette, for measuring out a volume of liquid

Volumetric flask, for making up a solution to a known volume

Figure 1.G Some volume measuring devices

Celsius, calibrated 0° to 100° between the freezing and boiling points of water

Fahrenheit, running 32° to 212° across that same interval

Kelvin, the SI scale, on which no temperature is negative

Figure 1.H Thermometers using different temperature scales

Measuring volume in the laboratory

- in the laboratory the volume of liquids or solutions can be measured by graduated cylinder, burette and pipette
- a volumetric flask is used to prepare a known volume of a solution

Density

- the two properties mass and volume are related as density = mass divided by volume, so density of a substance is its amount of mass per unit volume, and the SI unit of density is kg per m3, that is kg m-3
- this unit is quite large and a chemist often expresses density in g cm-3, where mass is expressed in gram and volume is expressed in cm3, and density of a substance tells us about how closely its particles are packed, so if density is more it means particles are more closely packed

1.3.7 Temperature, and 1.4 uncertainty in measurement

Three scales, and why numbers need care

- there are three common scales to measure temperature, degree celsius, degree fahrenheit and kelvin, of which K is the SI unit, and generally the thermometer with celsius scale is calibrated from 0° to 100°, where these two temperatures are the freezing point and the boiling point of water respectively, while the fahrenheit scale is represented between 32° to 212°
- the temperatures on the two scales are related to each other as F = nine fifths of C plus 32, and the kelvin scale is related to celsius scale as K = °C + 273.15
- it is interesting to note that temperature below 0 °C, that is negative values, are possible in Celsius scale but in Kelvin scale negative temperature is not possible
- many a time in the study of chemistry one has to deal with experimental data as well as theoretical calculations, and there are meaningful ways to handle the numbers conveniently and present the data realistically with certainty to the extent possible

Reference standard

The kilogram, and the search for a better one

- after defining a unit of measurement such as the kilogram or the metre, scientists agreed on reference standards that make it possible to calibrate all measuring devices, and for getting reliable measurements all devices such as metre sticks and analytical balances have been calibrated by their manufacturers to give correct readings, however each of these devices is standardised or calibrated against some reference
- the mass standard is the kilogram since 1889, and it has been defined as the mass of platinum-iridium, Pt-Ir, cylinder that is stored in an airtight jar at International Bureau of Weights and Measures in Sevres, France, Pt-Ir having been chosen for this standard because it is highly resistant to chemical attack and its mass will not change for an extremely long time
- scientists are in search of a new standard for mass, which is being attempted through accurate determination of Avogadro constant, work on it focusing on ways to measure accurately the number of atoms in a well-defined mass of sample, and one such method, which uses X-rays to determine the atomic density of a crystal of ultrapure silicon, has an accuracy of about 1 part in 106 but has not yet been adopted to serve as a standard, there being other methods but none of them presently adequate to replace the Pt-Ir cylinder

The metre, redefined twice

- the metre was originally defined as the length between two marks on a Pt-Ir bar kept at a temperature of 0°C, 273.15 K
- in 1960 the length of the metre was defined as 1.65076373 × 106 times the wavelength of light emitted by a krypton laser, and although this was a cumbersome number it preserved the length of the metre at its agreed value
- the metre was redefined in 1983 by CGPM as the length of path travelled by light in vacuum during a time interval of 1/299 792 458 of a second, and similar to the length and the mass there are reference standards for other physical quantities

1.4.1 Scientific notation

Numbers too large and too small to write out

- as chemistry is the study of atoms and molecules, which have extremely low masses and are present in extremely large numbers, a chemist has to deal with numbers as large as 602, 200,000,000,000,000,000,000 for the molecules of 2 g of hydrogen gas or as small as 0.000000000000000000000000166 g mass of a H atom, and other constants such as Planck's constant, speed of light and charges on particles involve numbers of the above magnitude
- it may look funny for a moment to write or count numbers involving so many zeros but it offers a real challenge to do simple mathematical operations of addition, subtraction, multiplication or division with such numbers, and this problem is solved by using scientific notation, that is exponential notation in which any number can be represented in the form N × 10n, where n is an exponent having positive or negative values and N is a number, called digit term, which varies between 1.000... and 9.999...
- thus we can write 232.508 as 2.32508 × 102 in scientific notation, and note that while writing it the decimal had to be moved to the left by two places and same is the exponent, 2, of 10 in the scientific notation
- similarly 0.00016 can be written as 1.6 × 10 -4, here the decimal has to be moved four places to the right and -4 is the exponent in the scientific notation

Multiplication and division

- these two operations follow the same rules which are there for exponential numbers, so the digit terms are multiplied or divided and the exponents are added or subtracted
- 5.6 × 105 times 6.9 × 108 gives 5.6 times 6.9 times 10 to the 5 plus 8, that is 38.64 × 1013, written as 3.864 × 1014
- 9.8 × 10-2 times 2.5 × 10-6 gives 24.50 × 10-8, written as 2.450 × 10-7, while 2.7 × 10-3 divided by 5.5 × 104 gives 0.4909 × 10-7, written as 4.909 × 10-8

Addition and subtraction

- for these two operations, first the numbers are written in such a way that they have the same exponent, and after that the coefficients, digit terms, are added or subtracted as the case may be
- thus for adding 6.65 × 104 and 8.95 × 103 the exponent is made same for both the numbers, giving 6.65 × 104 plus 0.895 × 104, which added is 7.545 × 104
- similarly the subtraction of 4.8 × 10-3 from 2.5 × 10-2 is done by writing the second as 0.48 × 10-2, giving 2.5 minus 0.48 all times 10-2, that is 2.02 × 10-2

1.4.2 Significant figures

Why a measurement is uncertain

- every experimental measurement has some amount of uncertainty associated with it because of limitation of measuring instrument and the skill of the person making the measurement, so the mass of an object obtained using a platform balance may come out to be 9.4g while on an analytical balance the mass obtained is 9.4213g, the second being slightly higher, therefore the digit 4 placed after decimal in the measurement by platform balance is uncertain
- the uncertainty in the experimental or the calculated values is indicated by mentioning the number of significant figures, which are meaningful digits known with certainty plus one which is estimated or uncertain, so if we write a result as 11.2 mL we say the 11 is certain and 2 is uncertain, and unless otherwise stated an uncertainty of plus or minus 1 in the last digit is always understood

The rules for counting them

- all non-zero digits are significant, for example in 285 cm there are three significant figures and in 0.25 mL there are two significant figures
- zeros preceding to first non-zero digit are not significant, such zero indicating the position of decimal point, thus 0.03 has one significant figure and 0.0052 has two
- zeros between two non-zero digits are significant, thus 2.005 has four significant figures
- zeros at the end or right of a number are significant provided they are on the right side of the decimal point, for example 0.200 g has three significant figures, but the terminal zeros are not significant if there is no decimal point, so 100 has only one significant figure while 100.0 has four, and such numbers are better represented in scientific notation, 100 being written as 1 × 102 for one significant figure, 1.0 × 102 for two and 1.00 × 102 for three

Exact numbers, and scientific notation

- counting the numbers of object, for example 2 balls or 20 eggs, have infinite significant figures as these are exact numbers and can be represented by writing infinite number of zeros after placing a decimal, that is 2 = 2.000000 or 20 = 20.000000
- in numbers written in scientific notation all digits are significant, so 4.01 × 102 has three significant figures and 8.256 × 10-3 has four

Precision is not accuracy

- precision refers to the closeness of various measurements for the same quantity, however accuracy is the agreement of a particular value to the true value of the result
- if the true value for a result is 2.00 g and student A reports 1.95 g and 1.93 g, these values are precise as they are close to each other but are not accurate, while student B obtains 1.94 g and 2.05 g, which are neither precise nor accurate, and student C reports 2.01 g and 1.99 g, which are both precise and accurate

Carrying significant figures through arithmetic

- in addition and subtraction the result cannot have more digits to the right of the decimal point than either of the original numbers, so adding 12.11, 18.0 and 1.012 gives 31.122, but 18.0 has only one digit after the decimal point and the result should be reported only up to one digit after the decimal point, which is 31.1
- in multiplication and division the result must be reported with no more significant figures as in the measurement with the few significant figures, so 2.5 × 1.25 = 3.125, but since 2.5 has two significant figures the result should not have more than two, thus it is 3.1

Rounding off

- if the rightmost digit to be removed is more than 5, the preceding number is increased by one, for example in 1.386, if we have to remove 6 we have to round it to 1.39
- if the rightmost digit to be removed is less than 5, the preceding number is not changed, for example 4.334, if 4 is to be removed then the result is rounded upto 4.33
- if the rightmost digit to be removed is 5, then the preceding number is not changed if it is an even number but it is increased by one if it is an odd number, so if 6.35 is to be rounded by removing 5 we have to increase 3 to 4 giving 6.4 as the result, however if 6.25 is to be rounded off it is rounded off to 6.2

1.4.3 Dimensional analysis

- often while calculating there is a need to convert units from one system to the other, and the method used to accomplish this is called factor label method or unit factor method or dimensional analysis
- a piece of metal is 3 inch, represented by in, long, and the question is what is its length in cm

Table 1.D Data to Illustrate Precision and Accuracy

	Measurements/g 1	Measurements/g 2	Average (g)
Student A	1.95	1.93	1.940
Student B	1.94	2.05	1.995
Student C	2.01	1.99	2.000

The unit factor, worked

- we know that 1 in = 2.54 cm, and from this equivalence we can write 2.54 cm over 1 in, which equals 1, and 1 in over 2.54 cm, which also equals 1, both of these being called unit factors, and if some number is multiplied by these unit factors, that is by 1, it will not be affected otherwise
- so the 3 in given above multiplied by the unit factor 2.54 cm over 1 in gives 3 times 2.54 cm = 7.62 cm, and the unit factor by which multiplication is to be done is the one which gives the desired units, the numerator having that part which is required in the desired result
- it should also be noted that units can be handled just like other numerical part, they can be cancelled, divided, multiplied and squared

Two more conversions

- a jug contains 2 L of milk, and to get its volume in m3 we use 1 L = 1000 cm3 and 1 m = 100 cm, cube the unit factor 1 m over 100 cm to get 1 m3 over 106 cm3, and multiply 2 × 1000 cm3 by it to obtain 2 × 10-3 m3
- to find how many seconds there are in 2 days, the unit factors 24 h over 1 day, 60 min over 1 h and 60 s over 1 min can be multiplied in series in one step only, giving 2 times 24 times 60 times 60 s = 172800 s

1.5 Laws of chemical combinations

Five basic laws, and the first of them

- the combination of elements to form compounds is governed by five basic laws
- the law of conservation of mass was put forth by Antoine Lavoisier, 1743-1794, in 1789, who performed careful experimental studies for combustion reactions and reached the conclusion that in all physical and chemical changes there is no net change in mass duting the process
- hence he reached the conclusion that matter can neither be created nor destroyed, which is called the Law of Conservation of Mass, and this law formed the basis for several later developments in chemistry
- infact this was the result of exact measurement of masses of reactants and products, and carefully planned experiments performed by Lavoisier

1.5.2 Law of definite proportions

- this law was given by a French chemist, Joseph Proust, 1754-1826, who stated that a given compound always contains exactly the same proportion of elements by weight
- Proust worked with two samples of cupric carbonate, one of which was of natural origin and the other was synthetic, and he found that the composition of elements present in it was same for both the samples, each being 51.35 per cent copper, 9.74 per cent carbon and 38.91 per cent oxygen
- thus he concluded that irrespective of the source a given compound always contains same elements combined together in the same proportion by mass, the validity of this law having been confirmed by various experiments, and it is sometimes also referred to as Law of Definite Composition

1.5.3 Law of multiple proportions

- this law was proposed by Dalton in 1803, and according to it, if two elements can combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in the ratio of small whole numbers
- for example hydrogen combines with oxygen to form two compounds, namely water and hydrogen peroxide, 2g of hydrogen with 16g of oxygen giving 18g of water, and 2g of hydrogen with 32g of oxygen giving 34g of hydrogen peroxide
- here the masses of oxygen, 16 g and 32 g, which combine with a fixed mass of hydrogen, 2g, bear a simple ratio, that is 16:32 or 1: 2

1.5.4 Gay Lussac's law of gaseous volumes

- this law was given by Gay Lussac in 1808, who observed that when gases combine or are produced in a chemical reaction they do so in a simple ratio by volume, provided all gases are at the same temperature and pressure
- thus 100 mL of hydrogen combine with 50 mL of oxygen to give 100 mL of water vapour, so the volumes of hydrogen and oxygen which combine bear a simple ratio of 2:1
- Gay Lussac's discovery of integer ratio in volume relationship is actually the law of definite proportions by volume, the law of definite proportions stated earlier having been with respect to mass, and Gay Lussac's law was explained properly by the work of Avogadro in 1811

1.5.5 Avogadro's law

- in 1811 Avogadro, 1776-1856, proposed that equal volumes of all gases at the same temperature and pressure should contain equal number of molecules, and he made a distinction between atoms and molecules which is quite understandable in present times
- if we consider again the reaction of hydrogen and oxygen to produce water, we see that two volumes of hydrogen combine with one volume of oxygen to give two volumes of water without leaving any unreacted oxygen, each box in the figure at the top of the next page containing equal number of molecules
- in fact Avogadro could explain the above result by considering the molecules to be polyatomic, and if hydrogen and oxygen were considered as diatomic as recognised now then the above results are easily understandable, however Dalton and others believed at that time that atoms of the same kind cannot combine

Two boxes of hydrogen and one of oxygen giving two boxes of water vapour

Every box holds the same number of molecules, which is Avogadro's proposal

No oxygen is left unreacted

Figure 1.I Two volumes of hydrogen react with one volume of oxygen to give two volumes of water vapour

The proposal that waited fifty years

- the belief of the time was that molecules of oxygen or hydrogen containing two atoms did not exist, and Avogadro's proposal was published in the French Journal de Physique, but in spite of being correct it did not gain much support
- after about 50 years, in 1860, the first international conference on chemistry was held in Karlsruhe, Germany, to resolve various ideas, and at the meeting Stanislao Cannizaro presented a sketch of a course of chemical philosophy, which emphasised on the importance of Avogadro's work

1.6 Dalton's atomic theory

- although the origin of the idea that matter is composed of small indivisible particles called a-tomio, meaning indivisible, dates back to the time of Democritus, a Greek Philosopher, 460-370 BC, it again started emerging as a result of several experimental studies which led to the laws mentioned above
- in 1808 Dalton, 1776-1884, published A New System of Chemical Philosophy, proposing that matter consists of indivisible atoms, that all atoms of a given element are identical in properties and mass while atoms of different elements differ in mass, that compounds form when atoms of different elements combine in a fixed ratio, and that chemical reactions reorganise atoms, which are neither created nor destroyed
- Dalton's theory could explain the laws of chemical combination, however it could not explain the laws of gaseous volumes, and it could not provide the reason for combining of atoms, which was answered later by other scientists

1.7 Atomic and molecular masses, and 1.7.1 atomic mass

Relative to carbon-12

- the mass of an atom is actually very-very small because atoms are extremely small, and today sophisticated techniques such as mass spectrometry determine atomic masses fairly accurately, but in the nineteenth century scientists could only determine the mass of one atom relative to another
- hydrogen, being the lightest atom, was arbitrarily assigned a mass of 1 without any units and other elements were assigned masses relative to it, however the present system agreed upon in 1961 is based on carbon-12, one of the isotopes of carbon, represented as 12C
- in this system 12C is assigned a mass of exactly 12 atomic mass unit, amu, and all other atoms are given masses relative to it, one atomic mass unit being defined as a mass exactly equal to one-twelfth of the mass of one carbon - 12 atom

A three-dimensional lattice of alternating sodium and chloride ions

Each Na+ has six Cl- around it, and each Cl- has six Na+

There is no single molecule to weigh, which is why a formula mass is used

Figure 1.J Packing of Na+ and Cl– ions in sodium chloride

The size of one unit

- 1 amu = 1.66056 × 10-24 g, and the mass of an atom of hydrogen is 1.6736 × 10-24 g, so in terms of amu the mass of hydrogen atom is 1.0078 amu, that is 1.0080 amu
- similarly the mass of oxygen - 16, 16O, atom would be 15.995 amu, and at present amu has been replaced by u, which is known as unified mass

1.7.2 Average atomic mass

- many naturally occurring elements exist as more than one isotope, and when we take into account the existence of these isotopes and their relative abundance, per cent occurrence, the average atomic mass of that element can be computed
- carbon has three isotopes, 12C with relative abundance 98.892 per cent and atomic mass 12, 13C with 1.108 per cent and 13.00335 amu, and 14C with 2 × 10-10 per cent and 14.00317 amu
- from the above data the average atomic mass of carbon comes out to be 12.011 u, and in the periodic table of elements the atomic masses mentioned for different elements actually represent their average atomic masses

1.7.3 Molecular mass

- molecular mass is the sum of atomic masses of the elements present in a molecule, and it is obtained by multiplying the atomic mass of each element by the number of its atoms and adding them together
- molecular mass of methane, CH4, which contains one carbon atom and four hydrogen atoms, is 12.011 u plus 4 times 1.008 u, that is 16.043 u
- similarly molecular mass of water, H2O, is 2 times the atomic mass of hydrogen plus 1 times the atomic mass of oxygen, that is 2 times 1.008 u plus 16.00 u = 18.02 u

1.7.4 Formula mass

- some substances, such as sodium chloride, do not contain discrete molecules as their constituent units, and in such compounds positive, sodium ion, and negative, chloride ion, entities are arranged in a three-dimensional structure
- it may be noted that in sodium chloride one Na+ ion is surrounded by six Cl- ion and vice-versa
- the formula, such as NaCl, is used to calculate the formula mass instead of molecular mass as in the solid state sodium chloride does not exist as a single entity

Equal counts and unequal masses, one mole of each substance set side by side

The mass of each is its atomic, molecular or formula mass expressed in grams

Figure 1.K One mole of various substances

Formula mass of NaCl, and Problem 1.1

- the formula mass of sodium chloride is the atomic mass of sodium plus the atomic mass of chlorine, that is 23.0 u + 35.5 u = 58.5 u
- Problem 1.1 asks for the molecular mass of a glucose, C6H12O6, molecule, and the solution is 6 times 12.011 u plus 12 times 1.008 u plus 6 times 16.00 u, that is 72.066 u plus 12.096 u plus 96.00 u = 180.162 u

1.8 Why a mole is needed

- atoms and molecules are extremely small in size and their numbers in even a small amount of any substance is really very large, so to handle such large numbers a unit of convenient magnitude is required
- just as we denote one dozen for 12 items, score for 20 items and gross for 144 items, we use the idea of mole to count entities at the microscopic level, that is atoms, molecules, particles, electrons and ions
- in SI system, mole, symbol mol, was introduced as seventh base quantity for the amount of a substance

What the mole is, and how large

- the mole, symbol mol, is the SI unit of amount of substance, and one mole contains exactly 6.02214076 × 1023 elementary entities, this number being the fixed numerical value of the Avogadro constant, NA, when expressed in the unit mol-1 and called the Avogadro number, while the amount of substance, symbol n, of a system is a measure of the number of specified elementary entities, an elementary entity being an atom, a molecule, an ion, an electron, any other particle or specified group of particles
- the mole of a substance always contains the same number of entities no matter what the substance may be, and in order to determine this number precisely the mass of a carbon-12 atom was determined by a mass spectrometer and found to be equal to 1.992648 × 10-23 g, so knowing that one mole of carbon weighs 12 g the number of atoms in it works out to 6.0221367 × 1023 atoms per mol
- to appreciate the largeness of this number let us write it with all zeroes without using any powers of ten, 602213670000000000000000, and hence so many entities constitute one mole of a particular substance, so 1 mol of hydrogen atoms is 6.022 × 1023 atoms, 1 mol of water molecules is 6.022 × 1023 water molecules, and 1 mol of sodium chloride is 6.022 × 1023 formula units of sodium chloride
- having defined the mole it is easier to know the mass of one mole of a substance, which in grams is called its molar mass and is numerically equal to atomic, molecular or formula mass in u, so the molar mass of water is 18.02 g mol-1 and that of sodium chloride is 58.5 g mol-1

1.9 Percentage composition

Why the percentage is asked for

- so far we were dealing with the number of entities present in a given sample, but many a time information regarding the percentage of a particular element present in a compound is required, and the first question to ask of an unknown or new compound is what is its formula or what are its constituents and in what ratio are they present
- for known compounds also such information provides a check whether the given sample contains the same percentage of elements as present in a pure sample, in other words one can check the purity of a given sample by analysing this data
- the mass per cent of an element is the mass of that element in the compound divided by the molar mass of the compound, all times 100

Two worked examples

- water contains hydrogen and oxygen, and with a molar mass of 18.02 g the mass per cent of hydrogen is 11.18 and the mass per cent of oxygen is 16.00 over 18.02 times 100, that is 88.79
- the molecular formula of ethanol is C2H5OH and its molar mass is 2 times 12.01 plus 6 times 1.008 plus 16.00, that is 46.068 g
- so the mass per cent of carbon is 24.02 g over 46.068 g times 100 = 52.14 per cent, of hydrogen is 6.048 g over 46.068 g times 100 = 13.13 per cent, and of oxygen is 16.00 g over 46.068 g times 100 = 34.73 per cent

1.9.1 Empirical formula for molecular formula

The two kinds of formula, and Problem 1.2

- an empirical formula represents the simplest whole number ratio of various atoms present in a compound, whereas the molecular formula shows the exact number of different types of atoms present in a molecule of a compound
- if the mass per cent of various elements present in a compound is known its empirical formula can be determined, and the molecular formula can further be obtained if the molar mass is known
- Problem 1.2 gives a compound containing 4.07 per cent hydrogen, 24.27 per cent carbon and 71.65 per cent chlorine, with a molar mass of 98.96 g, and asks for its empirical and molecular formulas
- step 1 is conversion of mass per cent to grams, and since we are having mass per cent it is convenient to use 100 g of the compound as the starting material, so in the 100 g sample 4.07g hydrogen, 24.27g carbon and 71.65g chlorine are present, while step 2 is to convert into number of moles of each element by dividing the masses obtained above by respective atomic masses, giving moles of hydrogen 4.07 g over 1.008 g = 4.04, moles of carbon 24.27 g over 12.01 g = 2.021, and moles of chlorine 71.65 g over 35.453 g = 2.021

Problem 1.2, steps 3 to 5

- step 3 is to divide each of the mole values obtained above by the smallest number amongst them, and since 2.021 is smallest, division by it gives a ratio of 2:1:1 for H:C:Cl, while in case the ratios are not whole numbers they may be converted into whole number by multiplying by the suitable coefficient
- step 4 is to write down the empirical formula by mentioning the numbers after writing the symbols of respective elements, so CH2Cl is the empirical formula of the above compound
- step 5 is writing the molecular formula, first determining the empirical formula mass by adding the atomic masses of various atoms present in the empirical formula, which for CH2Cl is 12.01 plus 2 times 1.008 plus 35.453 = 49.48 g, then dividing molar mass by empirical formula mass to get 2 = n, then multiplying empirical formula by n to get the molecular formula, so with empirical formula CH2Cl and n = 2 the molecular formula is C2H4Cl2

1.10 Stoichiometry and stoichiometric calculations

What a balanced equation tells you

- the word stoichiometry is derived from two Greek words, stoicheion meaning element and metron meaning measure, so stoichiometry deals with the calculation of masses, sometimes volumes also, of the reactants and the products involved in a chemical reaction
- for the combustion of methane the balanced equation is CH4 (g) + 2O2 (g) → CO2 (g) + 2 H2O (g), where methane and dioxygen are called reactants and carbon dioxide and water are called products, all of them gases here, which has been indicated by the letter (g) in the brackets next to its formula, while in case of solids and liquids (s) and (l) are written respectively
- the coefficients 2 for O2 and H2O are called stoichiometric coefficients, and similarly the coefficient for CH4 and CO2 is one in each case, they representing the number of molecules, and moles as well, taking part in the reaction or formed in the reaction

What it means, four ways

- one mole of CH4(g) reacts with two moles of O2(g) to give one mole of CO2(g) and two moles of H2O(g)
- one molecule of CH4 (g) reacts with 2 molecules of O2(g) to give one molecule of CO2(g) and 2 molecules of H2O(g)
- 22.7 L of CH4(g) reacts with 45.4 L of O2 (g) to give 22.7 L of CO2 (g) and 45.4 L of H2O(g)
- 16 g of CH4 (g) reacts with 2 times 32 g of O2 (g) to give 44 g of CO2 (g) and 2 times 18 g of H2O (g), and from these relationships the given data can be interconverted, mass through moles through number of molecules, using mass over volume = density

1.10.1 Limiting reagent

- many a time reactions are carried out with the amounts of reactants that are different than the amounts as required by a balanced chemical reaction, and in such situations one reactant is in more amount than the amount required by balanced chemical reaction

The reagent that runs out first

- the reactant which is present in the least amount gets consumed after sometime, and after that further reaction does not take place whatever be the amount of the other reactant
- hence the reactant which gets consumed first limits the amount of product formed and is therefore called the limiting reagent, and in performing stoichiometric calculations this aspect is also to be kept in mind

1.10.2 Reactions in solutions

- a majority of reactions in the laboratories are carried out in solutions, therefore it is important to understand as how the amount of substance is expressed when it is present in the solution
- the concentration of a solution, or the amount of substance present in its given volume, can be expressed as mass per cent or weight per cent (w/w %), mole fraction, molarity or molality, each of which is studied in detail over the next pages

Balancing a chemical equation

Metals, non-metals and oxygen

- according to the law of conservation of mass, a balanced chemical equation has the same number of atoms of each element on both sides of the equation, and many chemical equations can be balanced by trial and error
- 4 Fe(s) + 3O2(g) → 2Fe2O3(s) and 2 Mg(s) + O2(g) → 2MgO(s) are balanced, since there are same number of metal and oxygen atoms on each side of the equations, however P4(s) + O2 (g) → P4O10(s) is not balanced, phosphorus atoms being balanced but not the oxygen atoms
- to balance it we must place the coefficient 5 on the left of oxygen on the left side of the equation to balance the oxygen atoms appearing on the right side, giving P4(s) + 5O2(g) → P4O10(s)

Combustion of propane, in five steps

- step 1, write down the correct formulas of reactants and products, here propane and oxygen are reactants and carbon dioxide and water are products, giving the unbalanced equation C3H8(g) + O2(g) → CO2 (g) + H2O(l)
- step 2, balance the number of C atoms, and since 3 carbon atoms are in the reactant therefore three CO2 molecules are required on the right side, while step 3, balance the number of H atoms, on the left there are 8 hydrogen atoms in the reactants and each molecule of water has two hydrogen atoms, so four molecules of water will be required for eight hydrogen atoms on the right side
- step 4, balance the number of O atoms, there are 10 oxygen atoms on the right side, 3 times 2 = 6 in CO2 and 4 times 1 = 4 in water, therefore five O2 molecules are needed to supply the required 10 oxygen atoms, giving C3H8 (g) +5O2 (g) → 3CO2 (g) + 4H2O (l)
- step 5, verify that the number of atoms of each element is balanced in the final equation, the equation showing three carbon atoms, eight hydrogen atoms and 10 oxygen atoms on each side, and always remember that subscripts in formulas of reactants and products cannot be changed to balance an equation

Problem 1.3, water from methane

- the question asks for the amount of water in g produced by the combustion of 16 g of methane
- the balanced equation is CH4 (g) + 2O2 (g) → CO2 (g) + 2H2O (g), and 16 g of CH4 corresponds to one mole, while from the equation 1 mol of CH4 (g) gives 2 mol of H2O (g)
- 2 mol of water is 2 times 2 plus 16, that is 2 times 18 = 36 g, so multiplying 2 mol H2O by the unit factor 18 g H2O over 1 mol H2O gives 36 g H2O

Problem 1.4, methane for a given CO2

- the question asks how many moles of methane are required to produce 22g CO2 (g) after combustion
- according to the chemical equation CH4 (g) + 2O2 (g) → CO2 (g) + 2H2O (g), 44g CO2 (g) is obtained from 16 g CH4 (g), since 1 mol CO2(g) is obtained from 1 mol of CH4(g)
- the number of moles of CO2 (g) is 22 g times 1 mol over 44 g = 0.5 mol, hence 0.5 mol CO2 (g) would be obtained from 0.5 mol CH4 (g), or 0.5 mol of CH4 (g) would be required to produce 22 g CO2 (g)

Problem 1.5, ammonia, and which reagent limits it

- 50.0 kg of N2 (g) and 10.0 kg of H2 (g) are mixed to produce NH3 (g), and the question asks for the amount of NH3 (g) formed and the limiting reagent in this situation
- the number of moles of N2 is 50.0 kg times 1000 g per kg over 28.0 g per mol = 17.86 × 102 mol, and the number of moles of H2 is 10.00 kg times 1000 g per kg over 2.016 g per mol = 4.96 × 103 mol
- 1 mol N2 (g) requires 3 mol H2 (g) for the reaction, hence for 17.86 × 102 mol of N2 the moles of H2 (g) required would be 5.36 × 103 mol, but we have only 4.96 × 103 mol H2, hence dihydrogen is the limiting reagent in this case, so NH3(g) would be formed only from that amount of available dihydrogen
- since 3 mol H2(g) gives 2 mol NH3(g), 4.96 × 103 mol H2 (g) gives 3.30 × 103 mol NH3 (g), and if they are to be converted to grams, 1 mol NH3 (g) = 17.0 g NH3 (g), so the answer is 3.30 × 103 mol times 17.0 g per mol

The answer to Problem 1.5, and the first way to state a concentration

- 3.30 × 103 times 17 g NH3 (g) is 56.1 × 103 g NH3, that is 56.1 kg NH3
- the first way of expressing concentration is mass per cent, obtained by the relation printed below

Mole fraction

- mole fraction is the ratio of number of moles of a particular component to the total number of moles of the solution
- if a substance A dissolves in substance B and their number of moles are nA and nB respectively, then the mole fractions of A and B are given as printed above

Problem 1.6

- a solution is prepared by adding 2 g of a substance A to 18 g of water, and the question asks for the mass per cent of the solute
- the working, as printed in the book, is set out below

Molarity, and dilution

- molarity is the most widely used unit and is denoted by M, defined as the number of moles of the solute in 1 litre of the solution, so molarity is the number of moles of solute over the volume of solution in litres
- suppose we have 1 M solution of a substance, say NaOH, and we want to prepare a 0.2 M solution from it, 1 M NaOH meaning 1 mol of NaOH present in 1 litre of the solution, so for 0.2 M solution we require 0.2 moles of NaOH dissolved in 1 litre solution, hence we have to take that volume of 1M NaOH solution which contains 0.2 mol of NaOH and dilute the solution with water to 1 litre
- if 1 mol is present in 1L or 1000 mL solution then 0.2 mol is present in 1000 mL over 1 mol times 0.2 mol solution, that is 200 mL solution, so 200 mL of 1M NaOH are taken and enough water is added to dilute it to make it 1 litre
- in fact for such calculations a general formula, M1 × V1 = M2 × V2, where M and V are molarity and volume respectively, can be used, and note that the number of moles of solute was 0.2 in 200 mL and it has remained the same even after dilution, as we have changed just the amount of solvent and have not done anything with respect to NaOH, but keep in mind the concentration

Problem 1.7

- the question asks for the molarity of NaOH in the solution prepared by dissolving its 4 g in enough water to form 250 mL of the solution, and the working as printed is set out at the side
- note that molarity of a solution depends upon temperature because volume of a solution is temperature dependent

Molality, and Problem 1.8

- molality is defined as the number of moles of solute present in 1 kg of solvent and is denoted by m
- Problem 1.8 gives the density of 3 M solution of NaCl as 1.25 g mL-1 and asks for the molality, so with M = 3 mol L-1 the mass of NaCl in 1 L solution is 3 times 58.5 = 175.5 g, the mass of 1L solution is 1000 times 1.25 = 1250 g, and the mass of water in solution is 1250 -75.5 = 1074.5 g, giving molality 3 mol over 1.0745 kg = 2.79 m
- often in a chemistry laboratory a solution of a desired concentration is prepared by diluting a solution of known higher concentration, the solution of higher concentration being also known as stock solution, and note that the molality of a solution does not change with temperature since mass remains unaffected with temperature

Summary

What this unit covered, first part

- chemistry as we understand it today is not a very old discipline, people in ancient India already had the knowledge of many scientific phenomenon much before the advent of modern science and applied that knowledge in various walks of life
- the study of chemistry is very important as its domain encompasses every sphere of life, and chemists study the properties and structure of substances and the changes undergone by them, all substances containing matter, which can exist in three states, solid, liquid or gas, the constituent particles being held in different ways in these states and exhibiting their characteristic properties
- matter can also be classified into elements, compounds or mixtures, an element containing particles of only one type which may be atoms or molecules, compounds being formed where atoms of two or more elements combine in a fixed ratio to each other, and mixtures occurring widely, many of the substances present around us being mixtures
- when the properties of a substance are studied measurement is inherent, and the quantification of properties requires a system of measurement and units in which the quantities are to be expressed, many systems of measurement existing, of which the English system is one

Summary, continued

- the English and the Metric Systems are widely used, however the scientific community has agreed to have a uniform and common system throughout the world, which is abbreviated as SI units, International System of Units
- since measurements involve recording of data which are always associated with a certain amount of uncertainty, the proper handling of data obtained by measuring the quantities is very important, the measurements of quantities in chemistry being spread over a wide range of 10-31 to 10+23, hence a convenient system of expressing the numbers in scientific notation is used, the uncertainty being taken care of by specifying the number of significant figures in which the observations are reported, while dimensional analysis helps to express the measured quantities in different systems of units, so it is possible to interconvert the results from one system of units to another

Summary, concluded

- the combination of different atoms is governed by basic laws of chemical combination, these being the Law of Conservation of Mass, Law of Definite Proportions, Law of Multiple Proportions, Gay Lussac's Law of Gaseous Volumes and Avogadro Law, all of which led to Dalton's atomic theory, which states that atoms are building blocks of matter
- the atomic mass of an element is expressed relative to 12C isotope of carbon, which has an exact value of 12u, and usually the atomic mass used for an element is the average atomic mass obtained by taking into account the natural abundance of different isotopes of that element, while the molecular mass of a molecule is obtained by taking sum of the atomic masses of different atoms present in a molecule, and the molecular formula can be calculated by determining the mass per cent of different elements present in a compound and its molecular mass
- the number of atoms, molecules or any other particles present in a given system are expressed in the terms of Avogadro constant, 6.022 × 1023, known as 1 mol of the respective particles or entities
- chemical reactions represent the chemical changes undergone by different elements and compounds, a balanced chemical equation providing a lot of information, the coefficients indicating the molar ratios and the respective number of particles taking part in a particular reaction, while the quantitative study of the reactants required or the products formed is called stoichiometry, by which the amount of one or more reactants required to produce a particular amount of product can be determined and vice-versa, and the amount of substance present in a given volume of a solution is expressed in number of ways, mass per cent, mole fraction, molarity and molality

Exercises

Exercises 1.1 to 1.5

- 1.1 Calculate the molar mass of the following: (i) H2O (ii) CO2 (iii) CH4
- 1.2 Calculate the mass per cent of different elements present in sodium sulphate (Na2SO4).
- 1.3 Determine the empirical formula of an oxide of iron, which has 69.9% iron and 30.1% dioxygen by mass.
- 1.4 Calculate the amount of carbon dioxide that could be produced when (i) 1 mole of carbon is burnt in air. (ii) 1 mole of carbon is burnt in 16 g of dioxygen. (iii) 2 moles of carbon are burnt in 16 g of dioxygen.
- 1.5 Calculate the mass of sodium acetate (CH3COONa) required to make 500 mL of 0.375 molar aqueous solution. Molar mass of sodium acetate is 82.0245 g mol-1.

Exercises, continued

Exercises 1.6 to 1.10

- 1.6 Calculate the concentration of nitric acid in moles per litre in a sample which has a density, 1.41 g mL-1 and the mass per cent of nitric acid in it being 69%.
- 1.7 How much copper can be obtained from 100 g of copper sulphate (CuSO4)?
- 1.8 Determine the molecular formula of an oxide of iron, in which the mass per cent of iron and oxygen are 69.9 and 30.1, respectively.
- 1.9 Calculate the atomic mass (average) of chlorine using the following data: 35Cl, per cent natural abundance 75.77, molar mass 34.9689; 37Cl, per cent natural abundance 24.23, molar mass 36.9659.
- 1.10 In three moles of ethane (C2H6), calculate the following: (i) Number of moles of carbon atoms. (ii) Number of moles of hydrogen atoms. (iii) Number of molecules of ethane.

Exercises 1.11 to 1.15

- 1.11 What is the concentration of sugar (C12H22O11) in mol L-1 if its 20 g are dissolved in enough water to make a final volume up to 2L?
- 1.12 If the density of methanol is 0.793 kg L-1, what is its volume needed for making 2.5 L of its 0.25 M solution?
- 1.13 Pressure is determined as force per unit area of the surface. The SI unit of pressure, pascal is as shown below: 1Pa = 1N m-2. If mass of air at sea level is 1034 g cm-2, calculate the pressure in pascal.
- 1.14 What is the SI unit of mass? How is it defined?
- 1.15 Match the following prefixes with their multiples: (i) micro (ii) deca (iii) mega (iv) giga (v) femto, and the multiples 106, 109, 10-6, 10-15, 10.

Exercises 1.16 to 1.19

- 1.16 What do you mean by significant figures?
- 1.17 A sample of drinking water was found to be severely contaminated with chloroform, CHCl3, supposed to be carcinogenic in nature. The level of contamination was 15 ppm (by mass). (i) Express this in per cent by mass. (ii) Determine the molality of chloroform in the water sample.
- 1.18 Express the following in the scientific notation: (i) 0.0048 (ii) 234,000 (iii) 8008 (iv) 500.0 (v) 6.0012
- 1.19 How many significant figures are present in the following? (i) 0.0025 (ii) 208 (iii) 5005

Exercises, continued

Exercises 1.19 continued, 1.20 and 1.21

- 1.19, continued: (iv) 126,000 (v) 500.0 (vi) 2.0034
- 1.20 Round up the following upto three significant figures: (i) 34.216 (ii) 10.4107 (iii) 0.04597 (iv) 2808
- 1.21 The following data are obtained when dinitrogen and dioxygen react together to form different compounds: (i) 14 g dinitrogen with 16 g dioxygen (ii) 14 g with 32 g (iii) 28 g with 32 g (iv) 28 g with 80 g.
- 1.21 (a) Which law of chemical combination is obeyed by the above experimental data? Give its statement. (b) Fill in the blanks in the following conversions: (i) 1 km = ... mm = ... pm (ii) 1 mg = ... kg = ... ng (iii) 1 mL = ... L = ... dm3

Exercises 1.22 to 1.24

- 1.22 If the speed of light is 3.0 × 108 m s-1, calculate the distance covered by light in 2.00 ns.
- 1.23 In a reaction A + B2 gives AB2, identify the limiting reagent, if any, in the following reaction mixtures. (i) 300 atoms of A + 200 molecules of B (ii) 2 mol A + 3 mol B (iii) 100 atoms of A + 100 molecules of B (iv) 5 mol A + 2.5 mol B (v) 2.5 mol A + 5 mol B
- 1.24 Dinitrogen and dihydrogen react with each other to produce ammonia according to the following chemical equation: N2 (g) + H2 (g) gives 2NH3 (g). (i) Calculate the mass of ammonia produced if 2.00 × 103 g dinitrogen reacts with 1.00 × 103 g of dihydrogen. (ii) Will any of the two reactants remain unreacted? (iii) If yes, which one and what would be its mass?

Exercises 1.25 to 1.27

- 1.25 How are 0.50 mol Na2CO3 and 0.50 M Na2CO3 different?
- 1.26 If 10 volumes of dihydrogen gas reacts with five volumes of dioxygen gas, how many volumes of water vapour would be produced?
- 1.27 Convert the following into basic units: (i) 28.7 pm (ii) 15.15 pm (iii) 25365 mg

Exercises, concluded

Exercises 1.28 to 1.31

- 1.28 Which one of the following will have the largest number of atoms? (i) 1 g Au (s) (ii) 1 g Na (s) (iii) 1 g Li (s) (iv) 1 g of Cl_2(g)
- 1.29 Calculate the molarity of a solution of ethanol in water, in which the mole fraction of ethanol is 0.040 (assume the density of water to be one).
- 1.30 What will be the mass of one 12C atom in g?
- 1.31 How many significant figures should be present in the answer of the following calculations? (i) 0.02856 × 298.15 × 0.112 all over 0.5785 (ii) 5 × 5.364 (iii) 0.0125 + 0.7864 + 0.0215

Exercises 1.32 and 1.33

- 1.32 Use the data given in the following table to calculate the molar mass of naturally occuring argon isotopes: 36Ar, isotopic molar mass 35.96755 g mol-1, abundance 0.337%; 38Ar, 37.96272 g mol-1, 0.063%; 40Ar, 39.9624 g mol-1, 99.600%.
- 1.33 Calculate the number of atoms in each of the following (i) 52 moles of Ar (ii) 52 u of He (iii) 52 g of He.

Exercises 1.34 to 1.36

- 1.34 A welding fuel gas contains carbon and hydrogen only. Burning a small sample of it in oxygen gives 3.38 g carbon dioxide, 0.690 g of water and no other products. A volume of 10.0 L (measured at STP) of this welding gas is found to weigh 11.6 g. Calculate (i) empirical formula, (ii) molar mass of the gas, and (iii) molecular formula.
- 1.35 Calcium carbonate reacts with aqueous HCl to give $CaCl_2$ and CO_2 according to the reaction, $CaCO_3$ (s) + 2 HCl (aq) → $CaCl_2$ (aq) + CO_2(g) + H_2O(l). What mass of $CaCO_3$ is required to react completely with 25 mL of 0.75 M HCl?
- 1.36 Chlorine is prepared in the laboratory by treating manganese dioxide (MnO_2) with aqueous hydrochloric acid according to the reaction 4 HCl (aq) + MnO_2(s) → 2H_2O (l) + $MnCl_2$(aq) + Cl_2 (g). How many grams of HCl react with 5.0 g of manganese dioxide?